Investigating the Link between Trait Responsiveness to Verbal Suggestion and the Placebo Effect within the Predictive Processing Framework

Utkarsh Mishra[1], Ryan Parsons[2,3], Devin B. Terhune[4]

[1]IIT Delhi [2]Seattle Children's Research Institute [3]University of Washington [4]King's College London

Background & Motivation

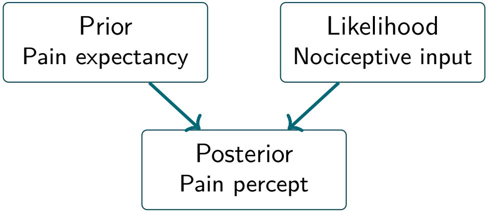
Pain as Bayesian Inference

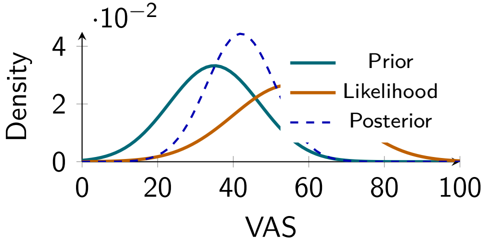
The final pain percept is a precision-weighted combination of:

- **Prior** – expected pain
- **Likelihood** – nociceptive input
- **Posterior** – experienced pain

Placebo hypoalgesia is a classic example of expectation-driven modulation of pain.

Trait **responsiveness to verbal suggestion** (REVS; BSS-C) predicts placebo responding in some paradigms, mediated by expectancy. The latent computational mechanisms remain unclear.





Methods

Experimental Paradigm

Re-analysis of Experiment 2 (Parsons et al., 2021), $N = 78$

- Double-blind, repeated-measures design
- Sham cream + verbal suggestion (Placebo vs Control regions)
- 64 trials of calibrated electrical stimulation
- Trial-by-trial VAS expectancy & pain ratings
- REVS measured with BSS-C (experimenter blind)

Bayesian Models

Prior = condition-specific expectancy
Likelihood = combined pain ratings
Posterior = product of two Gaussians

Three nested models (VBA toolbox)

- Prior-shift (p_{shift})
- Prior-precision ($p_{\text{precision}}$)
- Combined (both free)

Comparison via RFX Bayesian model selection.

Results, Planned Analyses & Implications

Exploratory Re-analysis (Exp. 1)

- Prior-precision model: **45/57** (79 %)
- Combined model: **12/57** (21 %)
- Combined-model participants showed larger placebo effects
- Average prior precision correlated with placebo magnitude ($r = .46$, $p < .001$)
- REVS did *not* predict placebo in Exp. 1 ($r = .11$, ns)

Planned Analyses (Exp. 2)

- Correlations: $p_{\text{precision}} \leftrightarrow$ placebo effect & BSS-C
- Bootstrap comparison of correlation strengths
- Partial correlations controlling for expectancy
- Mediation (SEM + bootstrap): does $p_{\text{precision}}$ mediate REVS $\rightarrow$ placebo?
- Group comparison by best-fitting model

Expected Outcomes & Implications

- Individual differences in placebo hypoalgesia are driven primarily by the *precision*